Respiratory volumes and capacities indicate the amount of air inspired and expired during normal respiration. Our respiratory system can be affected by pollutants, pathogens and other chemical substances found in air. Lung cancer and emphysema cannot be cured and these diseases are common among cigarette smokers.

People at higher level than the sea level are prompted to altitude sickness as the barometric pressure is low in those regions. Surfactant, emphysema, Asthma and Dead space have been discussed. During vigorous exercise the rate of respiration increases.

Activity

To test the presence of CO_2 in exhaled air Take two test tubes A and B with few mL of clear lime water. Blow exhaled air into A with a help of a straw and pass normal air into B with a help of a syringe for about 15 times and observe the changes that occur in the tubes A and B. The lime water (Calcium Hydroxide) in the test tube A turns milky.

Evaluation

1. Breathing is controlled by
   a. cerebrum
   b. medulla oblongata
   c. cerebellum
   d. pons

2. Intercostal muscles are found between the
   a. vertebral column b. sternum
   c. ribs d. glottis
3. The respiratory structures of insects are
   a. tracheal tubes b. gills
   c. green glands d. lungs
4. Asthma is caused due to
   a. inflammation of bronchus and bronchioles.
   b. inflammation of bronchione
   c. damage of diaphragm.
   d. infection of lungs
5. The Oxygen Dissociation Curve is
   a. sigmoid
   b. straight line
   c. curved
   d. rectangular hyperbola
6. The Tidal Volume of a normal person is
   a. 800 mL b. 1200 mL
   c. 500 mL d. 1100 – 1200 mL
7. During inspiration, the diaphragm
   a. expands.
   b. unchanged
   c. relaxes to become domed–shaped.
   d. contracts and flattens
8. CO_2 is transported through blood to lungs as
   a. carbonic acid
   b. oxyhaemoglobin
   c. carbamino haemoglobin
   d. carboxy haemoglobin
9. When 1500 mL air is in the lungs, it is called
   a. vital capacity
   b. tidal volume
   c. residual volume
   d. inspiratory reserve volume
10. Vital capacity is
    a. TV + IRV b. TV + ERV
    c. RV + ERV d. TV + TRV + ERV
11. After a long deep breath, we do not respire for some seconds due to

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